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Recent advances in electrical engineering: exploring graph neural networks for weather prediction in data-scarce environments

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Abstract

In regions like Nepal, characterized by diverse geography, missing weather data poses a significant challenge for traditional imputation methods. These methods often struggle to capture the complexities of dynamic environments adequately. To overcome this challenge, our study explores the application of graph neural networks for weather prediction in data-scarce environments. Our approach entails the development of specialized models tailored to accommodate the non-Euclidean topology inherent in weather data. This framework encompasses preprocessing, graph representation, feature selection, and imputation techniques to predict missing atmospheric variables. The adaptability of our models to intricate geography ensures more precise representations of weather conditions. Our research demonstrates the efficacy of these models through rigorous testing on a substantial dataset spanning four decades since 1981. By harnessing state-of-the-art graph neural network technology, our study aims to address existing gaps in weather data prediction, leading to improved historical weather forecasting accuracy. Ultimately, this advancement contributes to enhanced meteorological understanding and forecasting precision in data-scarce regions.

1. Introduction

In this global climate change scenario, the importance of accurate weather predictions is increasing. An abnormal weather condition, such as cold weather, heavy snow, heavy rain, and drought, occur more frequently worldwide. Weather change has a direct negative impact on daily quality of life, agriculture systems [1, 2], crop management, energy utilization [3], image denoising [4], industries [5], the tourism industry [6], transportation and logistics [7]. In the global scenario of agricultural landscape, weather-dependent decisions determine crop yields and food production. Similarly, energy grids and utility systems are highly sensitive to weather fluctuations. Inaccurate forecasts can strain these systems, culminating in costly outages, infrastructure damage, and disruptions in energy supply to homes and industries. Moreover, the transportation industry hinges on timely and precise weather information for route planning, safety precautions, and minimizing disruptions due to adverse weather conditions.

There is a range of techniques and approaches for predicting atmospheric variables using various machine learning and statistical methods. Existing weather forecasting models are based on physics and use supercomputers to evolve the atmosphere in the future. To minimize this computational cost, several deep-learning techniques have been proposed over the last few decades to forecast meteorological information. Some recent methods are heatwave prediction [8], air temperature forecasting [9], high-resolution prediction of spatiotemporal distribution of air pollution [10], precipitation prediction at a high resolution up to 12 hours ahead [11], and rainfall prediction models [12–14]. GNN-based methods have demonstrated exceptional performance in handling non-Euclidean weather data [15, 16]. Recently, GRAST [17] and HiSTGNN [18] have introduced GNNs designed for multi-station weather forecasting, addressing the need for enhanced modeling of

interactions among meteorological variables from various stations. Besides forecasting the future, sometimes it is essential to compute the missing atmospheric variable using other environmental factors. The specific variable could be missed due to the absence or malfunction of the measuring device.

Here we present a neural network to predict the current value of a specific atmospheric variable using other environmental factors. Probabilistic forecasts predict the likelihood of weather conditions at a given time and location. We proposed a Graph Neural Network (GNN) to classify the range of core atmospheric variables such as rainfall, temperature, humidity, and air pressure using other correlated environmental variables. For the atmospheric variables, a probabilistic outcome answers the question, “What is the range of probability of a given variable occurring at a location and time based on other correlated environmental factors?”

In developing countries, weather observations often include missing values, especially in diverse regions like Nepal, where the plains, hills, and Himalayas create distinct climatic zones within short distances. These gaps, caused by equipment malfunctions or human error, necessitate weather imputation, which estimates missing data using techniques such as graph neural networks [19–21], regression trees, artificial neural networks [22], rain gauge estimation [23], spatial-correlation approaches [24], statistical imputation with quality control measures [25], and machine learning [26, 27]. Recent methods like ARIMA [20], LSTM [19], BiLSTM [28], sequential imputation [29], dynamic time warping [30], and edge cutting [31] have further advanced this field. Weather imputation is crucial for maintaining continuous, reliable datasets and supporting decision-making in weather-sensitive activities [45]. However, methods like Numerical Weather Prediction (NWP) [32] and interpolation [33] face challenges, including capturing complex weather phenomena, handling sparse data, and managing the resource demands of these techniques [34].

In this work, we introduce ImputeCast, a groundbreaking machine learning paradigm designed to tackle the dual challenge of imputing partially missing attributes and predicting the range of entirely missing features. Leveraging two distinct approaches, graph diffusion and graph neural networks (GNN), the method operates on a systematically designed graph topology to enhance imputation accuracy. The graph diffusion technique exploits information propagation across interconnected nodes, while the GNN approach harnesses neural networks to capture intricate patterns in graph-structured data.

The major contributions of this paper are as follows:

- We introduce a novel weather data imputation method that opens up new research avenues.
- Our approach leverages Graph Neural Network (GNN) technology to predict missing atmospheric variables by utilizing other environmental features. We outline a comprehensive GNN processing pipeline, encompassing data preparation, feature selection, and neural network optimization.
- This study presents a novel GNN architecture, specifically the GraphSAGE network, tailored for predicting missing weather data attributes, including rainfall, temperature, humidity, and atmospheric pressure, using available weather information. The GNN demonstrates proficiency in handling all four atmospheric variables while achieving a commendable level of accuracy.
- We introduced a feature selection method based on Pearson's correlation coefficient that found comparatively better than the conventional approach.

2. Materials and methods

2.1. Dataset and preprocessing

Nepal's geographical landscapes exhibit a remarkable diversity in weather patterns and climate fluctuations. Spanning from the lowlands of Kechana Kalan, located at a modest elevation of 60 meters above sea level, to the awe-inspiring heights of Mount Everest, the world's tallest peak towering at an astonishing 8,848 meters, the country encompasses an astounding spectrum of altitudes. This geographical diversity encompasses an extraordinary range of climatic nuances and meteorological phenomena. Within this work, we utilized historical weather data sourced from 62 weather stations strategically distributed across diverse geographical regions within Nepal. This dataset covers a time span of 39 years, encompassing daily weather records from January 1, 1981, through December 31, 2019. Remarkably, this dataset encompasses more than 883K data samples, making it the most extensive and publicly available meteorological dataset [35, 36] within our knowledge. The dataset comprised of several attributes, namely PRECTOT: Precipitation (mm/day) PS: Surface Pressure (kPa), QV2M: Specific Humidity at 2 Meters (g/kg), RH2M: Relative Humidity at 2 Meters (%), T2M: Temperature at 2 Meters (C), T2MWET: Wet Bulb Temperature at 2 Meters (C), T2M_MAX: Maximum Temperature at 2 Meters (C), T2M_MIN: Minimum Temperature at 2 Meters (C), T2M_RANGE: Temperature Range at 2 Meters (C), TS: Earth Skin Temperature (C), WS10M: Wind Speed at 10 Meters (m/s), WS10M_MAX: Maximum Wind Speed

at 10 Meters (m/s), WS10M_MIN: Minimum Wind Speed at 10 Meters (m/s), WS10M_RANGE: Wind Speed Range at 10 Meters (m/s), WS50M: Wind Speed at 50 Meters (m/s), WS50M_MAX: Maximum Wind Speed at 50 Meters (m/s), WS50M_MIN: Minimum Wind Speed at 50 Meters (m/s), WS50M_RANGE: Wind Speed Range at 50 Meters (m/s).

The dataset we used contains no missing values initially. However, we hypothesize that missing values are a common occurrence in many datasets. To test the efficacy of our model, we applied a random filter to deliberately remove portions of the available data. This approach allows us to simulate missing data scenarios and evaluate how well the model performs under these conditions.

Following the dataset declaration, we proceed by representing the 62 weather stations as individual nodes within a graph. These nodes are connected by edges, which are established based on similarities in altitude and proximity to neighboring stations. We prepared the data to address a classification problem involving four atmospheric variables, each with its specific prediction range. Rainfall is classified into two categories, either present or absent, while temperature is segmented into six distinct classes ranging from freezing to very hot. Similarly, humidity is categorized into four classes—dry, moderate, humid, and very humid—while pressure is stratified into four levels: very low, low, moderate, and high. In the dataset, the maximum temperature recorded is 36°C. Accordingly, we categorized temperatures into the following ranges: below 0°C as freezing, 0–10°C as cold, 10–18°C as cool, 18–25°C as moderate, 25–30°C as hot, and above 30°C as very hot. Similarly, humidity is categorized into ranges of less than 40%, 40–60%, 60–80%, and above 80%. For pressure, the categories are less than 70 kPa, 70–80 kPa, 80–90 kPa, and 90–100 kPa. While the categorization process may seem random, it is purposeful and not arbitrary. Several studies underscore the importance of considering a broader spectrum of weather variables. To understand the dynamics of vector-borne diseases [37, 38], diseases on plant and vegetables [39], crop production prediction [40], minimizing extensive use of limited resources [41] utilities of above categories is crucial.

2.2. Graph representation of weather data

A graph is a kind of non-Euclidean data structure which shows the relationship between entities; such entities are known as the nodes and relationships are the edges of the graph. Formally, a graph can be defined as $G = (V, E)$ where $V = \{v_1, v_2, \dots, v_n\}$ is the set of n vertices and $E = \{e_1, e_2, \dots, e_m\}$ is the set of m edges. If the edges on the graph are directed from one node to another, then the graph formed is the directed graph. Also, if the strengths of the relationships between nodes are not the same, such graphs are weighted graphs. The graph defined above is an undirected and unweighted graph. In this weather imputation and missing variable prediction task, we used weighted undirected graph, which is mathematically represented as $G = (V, E, W)$ where V is the set of vertices, E is the set of edges, and W is a matrix of size $|V| \times |V|$ representing the strength of edges. In a graph signal processing task, a graph signal (graph feature) is a real mapping defined on the graph vertices V and edge weight (a real mapping on $V \times V$) is the connection strength of the vertices of the graph. The matrix form of the edge weight is the weighted adjacency matrix W , $D \in \mathbb{R}^{|V| \times |V|}$, $D_{ii} = \sum_j w_{ij}$ is the degree matrix of W , $\tilde{W} = D^{-\frac{1}{2}} W D^{-\frac{1}{2}}$ is the normalized weighted adjacency matrix and $\Delta = I - \tilde{W}$ is the graph Laplacian matrix.

In our approach, we conceptualize these weather stations as the nodes (vertices) of a graph, and the connections between these nodes represent the edges. These edges facilitate the exchange of information between the nodes, and weights are assigned to them to quantify the strength of the connection between any two nodes. Importantly, our edge formulation takes into account both spatial and temporal aspects. By considering both spatial relationships and temporal order in the edges, our approach provides a more comprehensive framework for analyzing and understanding meteorological data, allowing us to capture both the physical proximity of weather stations and the chronological sequence of events in our analysis.

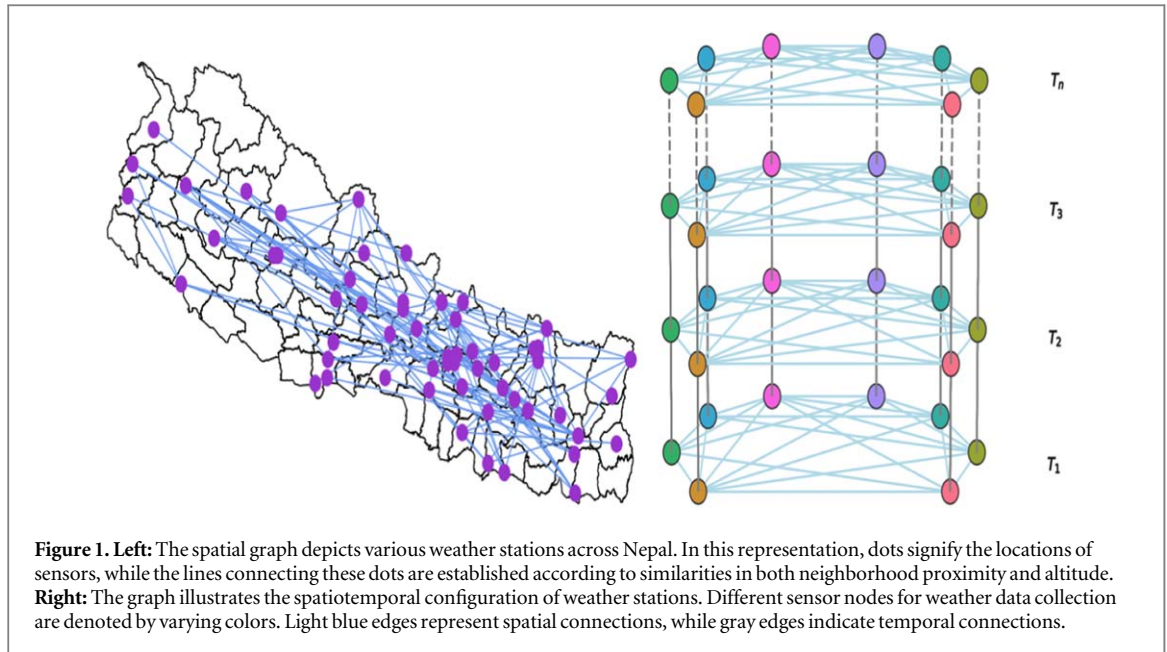
2.2.1. Spatial edge construction

Spatial edges represent relationships between nodes based on spatial proximity or physical distance. These edges form when nodes are physically close or share spatial traits, such as altitude. The altitude-based edges align with the hypothesis that locations at similar altitudes exhibit similar weather, including temperature variations due to the lapse rate. The degree of similarity between nodes is determined by this altitude-based criterion

$$S_{ij} = \frac{1}{|A(i) - A(j)|}$$

where $A(i)$ is the altitude of node i and S_{ij} is the similarity score between node i and node j and is the edge weight for edge (i, j) .

Inspired from the concept of neutrosophic relation [42], neighborhood-based edges are created based on the proximity principle, with edge weight determined by



$$w_{ij} = \frac{1}{g(i, j)}$$

where $g(i, j)$ is the geodesic distance between node i and j .

In both scenarios, a threshold is applied to maintain sparsity, and when both spatial methods establish connections between nodes, the edge weight is the maximum of S_{ij} and w_{ij} .

2.2.2. Temporal edge construction

Temporal edges establish time-based relationships between nodes, connecting data from the same station at consecutive points for enhanced temporal dependency and its weight t_{ij} is considered to be one.

The weather data we have used is the temporal data, so to leverage the temporal information, spatial graphs for every temporal point $t = 1, 2, \dots, T$ are stacked and consecutive temporal points on the same station are connected by an edge.

The full lines in the graph indicate the spatial edges and the dotted lines indicate temporal edges. Therefore, the graph obtained, called weather imputation graph (WIG), will have $n \times T$ nodes where n is the number of weather stations and T is the length of temporal points. Furthermore, the size of the WIG is $|E| = |ES| + |ET|$, where $|ET| = n \times (T - 1)$ is the number of temporal edges and $|ES|$ is the number of spatial edges which is $T \times$ the number of edges on a spatial graph at a temporal point. Figure 1 gives a brief overview of the spatio-temporal graph representing the weather stations and its temporal form.

2.3. Graph weather feature imputation

Graph-based weather feature imputation stands as a central focus of this research, where the task involves filling in the missing weather features X_m using the available features X_h . We employed the graph-based heat diffusion method [43] for weather data imputation. The imputation technique is based on the prior that weather features are similar in nearby geographical locations, areas of similar altitude, and at consecutive time intervals.

This can be quantified with the help of graph Dirichlet energy

$$E_G(f) = \frac{1}{2} f^T \Delta f = \frac{1}{2} \sum_{i,j} \tilde{w}_{ij} (f_i - f_j)^2 \quad (1)$$

where $\tilde{w}_{ij}$ is the individual entries of normalized weighted adjacency matrix and f_i if the features at node i . The Dirichlet energy (1) is used to measure the smoothness of weather data stored on the nodes on the weather imputation graph. That means the neighboring weather data encoded by weather imputation graph structure are similar. Thus, we will learn missing weather data from its neighboring nodes.

In the WIG, the total vertices V can be partitioned as $V = V_h \cup V_m$, where V_h is the set of nodes i.e. weather stations where the features are present at hand and V_m is the set of weather stations in which weather feature is missing. Then weather feature can be written as $f = \begin{bmatrix} f_h \\ f_m \end{bmatrix}$ where f_h is the weather feature at hand and f_m is the missing weather feature. Furthermore, the weighted adjacency matrix and graph Laplacian are respectively

$$W = \begin{bmatrix} W_{hh} & W_{hm} \\ W_{mh} & W_{mm} \end{bmatrix} \quad (2)$$

and

$$\Delta = \begin{bmatrix} \Delta_{hh} & \Delta_{hm} \\ \Delta_{mh} & \Delta_{mm} \end{bmatrix} \quad (3)$$

The weather feature imputation means we are attempting to find the missing weather feature f_m with the help of feature at hand f_h . We do so by message passing scheme in the graph via weighted normalized adjacency matrix until graph Dirichlet energy $E_G(f)$ is minimized. The closed form solution of the minimizer of Dirichlet energy is possible [43] but lacks computational efficiency, so it is better to use gradient flow

$$\dot{f} = -\nabla E_G(f)$$

as graph heat diffusion differential equation with initial condition as

$$f(0) = \begin{bmatrix} f_h \\ f_m(0) \end{bmatrix}$$

and boundary condition as the feature at hand $f_h(t) = f_h$. At the same time, gradient flow of feature at hand f_h should not be changing over time, so

$$\dot{f}_h = 0$$

With this and above initial and boundary conditions, change can only be applied to missing weather feature f_m and can be modeled as

$$\dot{f}_m(t) = -(\Delta_{mh}f_h + \Delta_{mm}f_m(t)) \quad (4)$$

Euler iterative scheme [43] for solving this is combined as

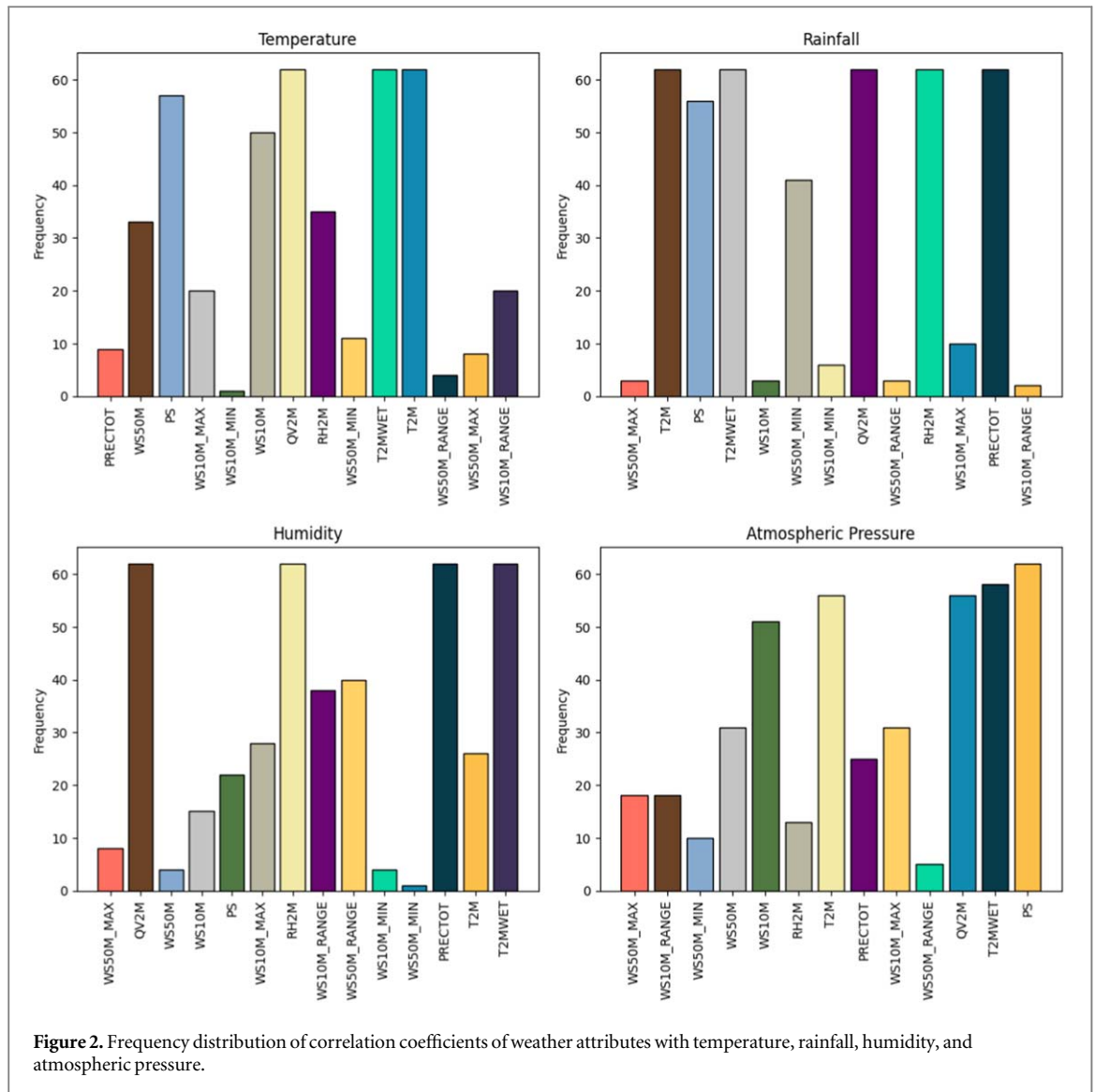
$$f^{(i)} = \begin{bmatrix} I & 0 \\ \tilde{W}_{mh} & \tilde{W}_{mm} \end{bmatrix} f^{(i-1)}; i = 1, 2, 3, \dots \quad (5)$$

which is the required weather data imputation scheme and it update only missing weather feature f_m .

3. Node feature selection method

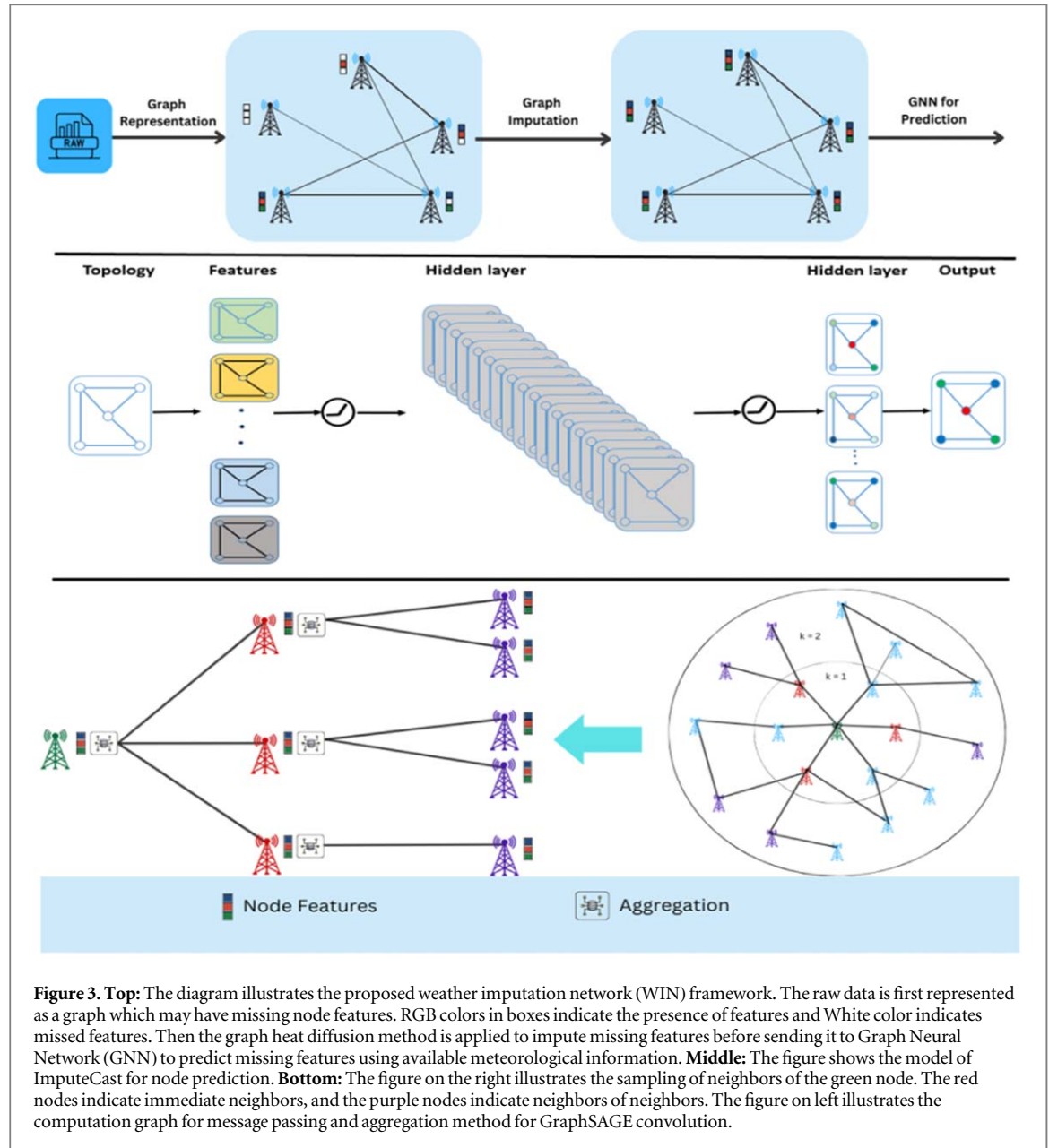
In the context of graph-based weather data processing, the careful choice of node features holds the potential to significantly reduce computational costs and enhance model efficiency. Feature selection reduces dimensionality, accelerates model training, and enhances interpretability. This process not only improves the accuracy of predictions, benefiting weather forecasting and spatial data analysis, but also filters out noise and ensures that computational resources are used efficiently.

In the study [44], Genetic algorithm (GA) emerged as a more efficient method than Wrapper approaches for rainfall prediction. Malathi et al, malathi2020feature conducted a comprehensive analysis of various weather feature selection strategies, including Information Gain (IG), Chi-Square (CS), Symmetric Uncertainty (SU), Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), and GA, concluding that the selection of these methods should be based on factors such as simplicity, stability, feature reduction, accuracy, and computational efficiency. The correlation based feature selection [46] was used for precipitation type prediction. However, it's essential to note that many of these methods require substantial domain knowledge and, in some instances, significant computational resources. In response, we have developed a straightforward yet highly effective feature selection strategy using a statistical approach—Pearson's correlation coefficient. This metric allowed us to quantify the degree of linear association between each feature and the specific meteorological parameter we sought to predict. The detail description can be found in algorithm 1.



Algorithm 1 Input:

- **Raw Data:** Different temporal weather information of different locations
- **x:** Target variable
- **Top Number:** Number of top features to select (e.g., 6)
- **Top Features:** List of the top 6 feature names by frequency
- **Output:** Initialize an empty list **feature_list** to store feature names.
- Find **locations** from **Raw Data**
- For each **location** in **locations**, do the following:
 - a. Calculate features for the current location: $f_1, f_2, \dots, f_n = \text{calculate_features}(\text{location})$
 - b. Calculate feature-target correlations: $r_1, r_2, \dots, r_n = \text{calculate_correlations}([f_1, f_2, \dots, f_n], x)$
 - c. Find the top num_top features with highest correlations: **top_features** = **find_top_features**($[f_1, f_2, \dots, f_n], [r_1, r_2, \dots, r_n], \text{num_top}$)
 - d. Add the top correlated feature names to **feature_list**: **feature_list.extend(top_features)**
- Create a dictionary **feature_count** to store feature frequency counts.



- For each feature in `feature_list`, do the following:
 - a. If the feature is not in `feature_count`, add it with a count of 1.
 - b. If the feature is already in `feature_count`, increment its count by 1.
- Sort `feature_count` by frequency in descending order.
- Select the top 6 feature names based on frequency: `top_features = top_n_keys(feature_count, num_top)`
- Return `top_features`, which contains the top 6 feature names by frequency.

The initial dataset contains weather information from 62 weather stations in Nepal, comprising 21 different features. For each of the four prediction tasks (rainfall, temperature, humidity, and atmospheric pressure), we computed Pearson's correlation coefficients between all available features and the respective meteorological parameter. The aim is to reconstruct the data with minimal information. Therefore, we selected the top six features using Algorithm-I, and this number serves as a hyper-parameter. Through the utilization of Algorithm-I, the six most pertinent features have been identified for various weather aspects: temperature (WS10M, RH2M, PS, T2MWET, WS50M and QV2M), rainfall (RH2M, T2M, PS, T2MWET, WS50M_MIN and QV2M), Humidity (PRECTOT, T2M, T2MWET, WS10M_RANGE, WS50M_RANGE and WS10M_MAX), and

Table 1. Summary of weather prediction.

Temperature Prediction												
Class	0%		30%		60%		90%		F1	Pre	Rec	F1
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec				
0	0.95	0.94	0.94	0.92	0.90	0.91	0.89	0.86	0.87	0.81	0.77	0.79
1	0.93	0.93	0.93	0.88	0.89	0.88	0.83	0.84	0.84	0.74	0.73	0.74
2	0.93	0.92	0.93	0.87	0.87	0.87	0.81	0.81	0.81	0.70	0.74	0.72
3	0.91	0.91	0.91	0.86	0.84	0.85	0.79	0.79	0.79	0.70	0.67	0.69
4	0.87	0.89	0.88	0.81	0.85	0.83	0.75	0.79	0.77	0.65	0.72	0.68
5	0.85	0.81	0.83	0.80	0.68	0.74	0.73	0.57	0.64	0.65	0.35	0.45

Humidity Prediction												
Class	0%		30%		60%		90%		F1	Pre	Rec	F1
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec				
0	0.84	0.83	0.84	0.80	0.71	0.75	0.74	0.58	0.65	0.63	0.41	0.49
1	0.85	0.88	0.86	0.77	0.84	0.81	0.70	0.81	0.75	0.59	0.78	0.67
2	0.75	0.70	0.73	0.65	0.61	0.63	0.56	0.52	0.54	0.46	0.32	0.38
3	0.87	0.87	0.87	0.82	0.80	0.81	0.78	0.73	0.75	0.65	0.62	0.63

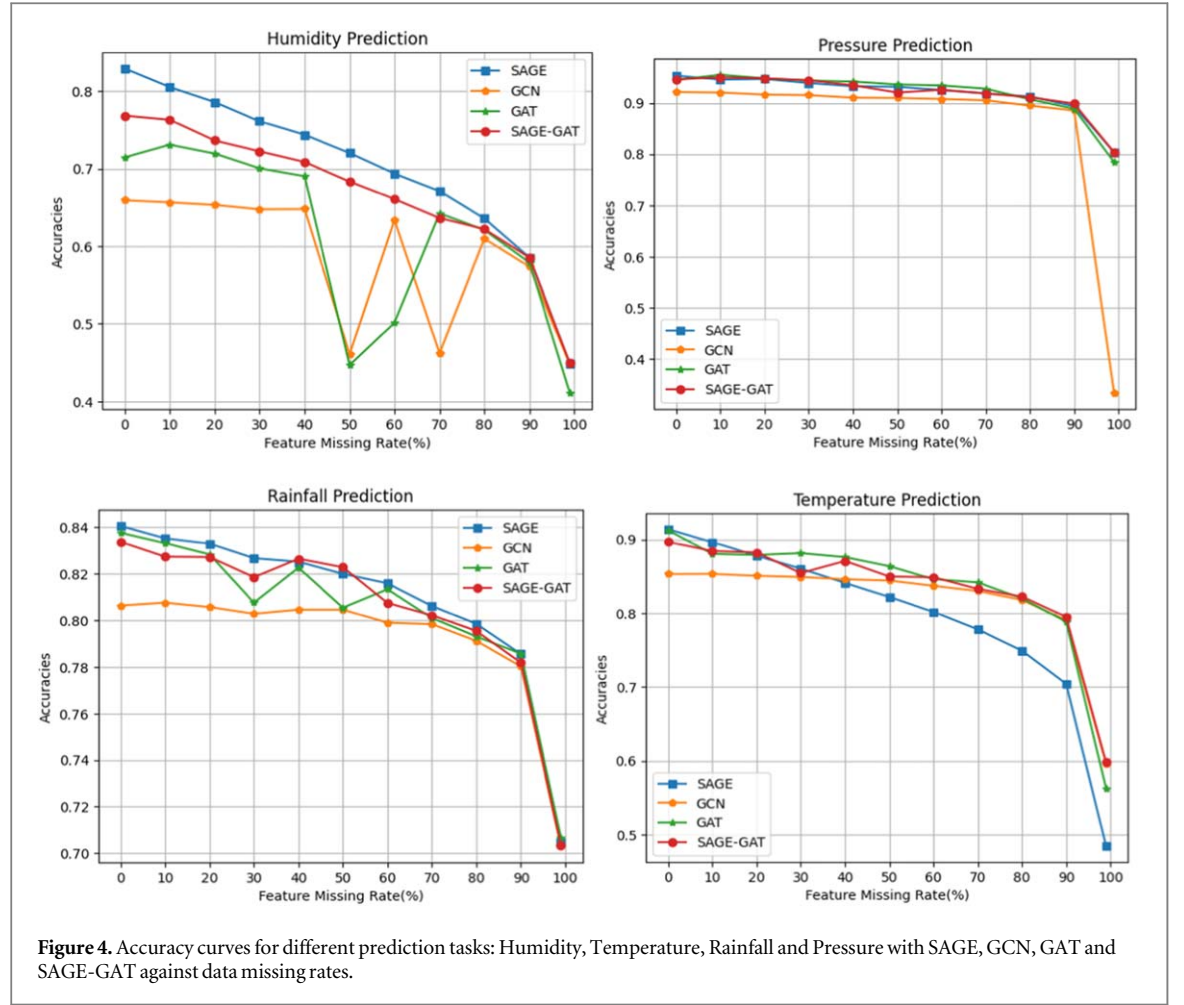
Pressure Prediction												
Class	0%		30%		60%		90%		F1	Pre	Rec	F1
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec				
0	0.97	0.97	0.97	0.96	0.97	0.96	0.96	0.96	0.96	0.94	0.93	0.93
1	0.93	0.93	0.93	0.91	0.92	0.91	0.89	0.90	0.89	0.83	0.87	0.85
2	0.94	0.92	0.93	0.92	0.89	0.90	0.90	0.86	0.88	0.86	0.80	0.83
3	0.98	0.99	0.99	0.97	0.98	0.98	0.96	0.98	0.97	0.95	0.96	0.97

Rainfall Prediction												
Class	0%		30%		60%		90%		F1	Pre	Rec	F1
	Pre	Rec	F1	Pre	Rec	F1	Pre	Rec				
0	0.90	0.90	0.90	0.88	0.89	0.88	0.86	0.88	0.87	0.81	0.88	0.84
1	0.78	0.78	0.78	0.75	0.73	0.74	0.72	0.67	0.70	0.67	0.56	0.61

Atmospheric Pressure (T2MWET, WS50M, WS50M_MAX, WS10M, QV2M and T2M). The figure 2 provides a visual representation of the frequency distribution of the selected features.

4. Network architecture

In the work for missing atmospheric variable prediction, we employed several variants of Graph Neural Networks (GNN). These encompass Graph Convolutional Networks (GCN), renowned for their ability to capture local graph structure; GraphSAGE (Graph Sample and Aggregated Embeddings) [47], which adeptly samples and aggregates information from neighboring nodes; Graph Attention Networks (GAT), leveraging attention mechanisms to weigh node importance; and a hybrid approach known as SAGE-GAT, combining both sampling and attention-based aggregation techniques. These distinct networks exhibit the ability to capture both local and global dependencies within the atmospheric data graph, contributing to precise predictions of missing atmospheric variables. In the context of predicting missing weather data variables, GraphSAGE consistently outperforms other classifiers in all scenarios. Consequently, our subsequent discussion predominantly focuses on the SAGE neural network for classifying the range of missing atmospheric variables. GraphSAGE generates node embedding in a graph by sampling neighboring nodes and aggregating their features. It uses a fixed number of neighboring nodes (neighbor sampling) for each node in the graph, ensuring consistent computational and memory requirements regardless of graph size. It then combines these neighboring node features to create embedding for the target node. These embedding are learned and optimized during training to capture meaningful representations of graph nodes in a lower-dimensional space. Mathematically, let h_v^{l-1} be embedding of a node v at layer $l - 1$, then embedding of the node v at layer l is given by



$$h_v^l = \sigma([W_l \cdot \text{AGG}(\{h_u^{l-1}, \forall u \in \mathcal{N}(v)\}) + B_l h_v^{l-1}]) \quad (6)$$

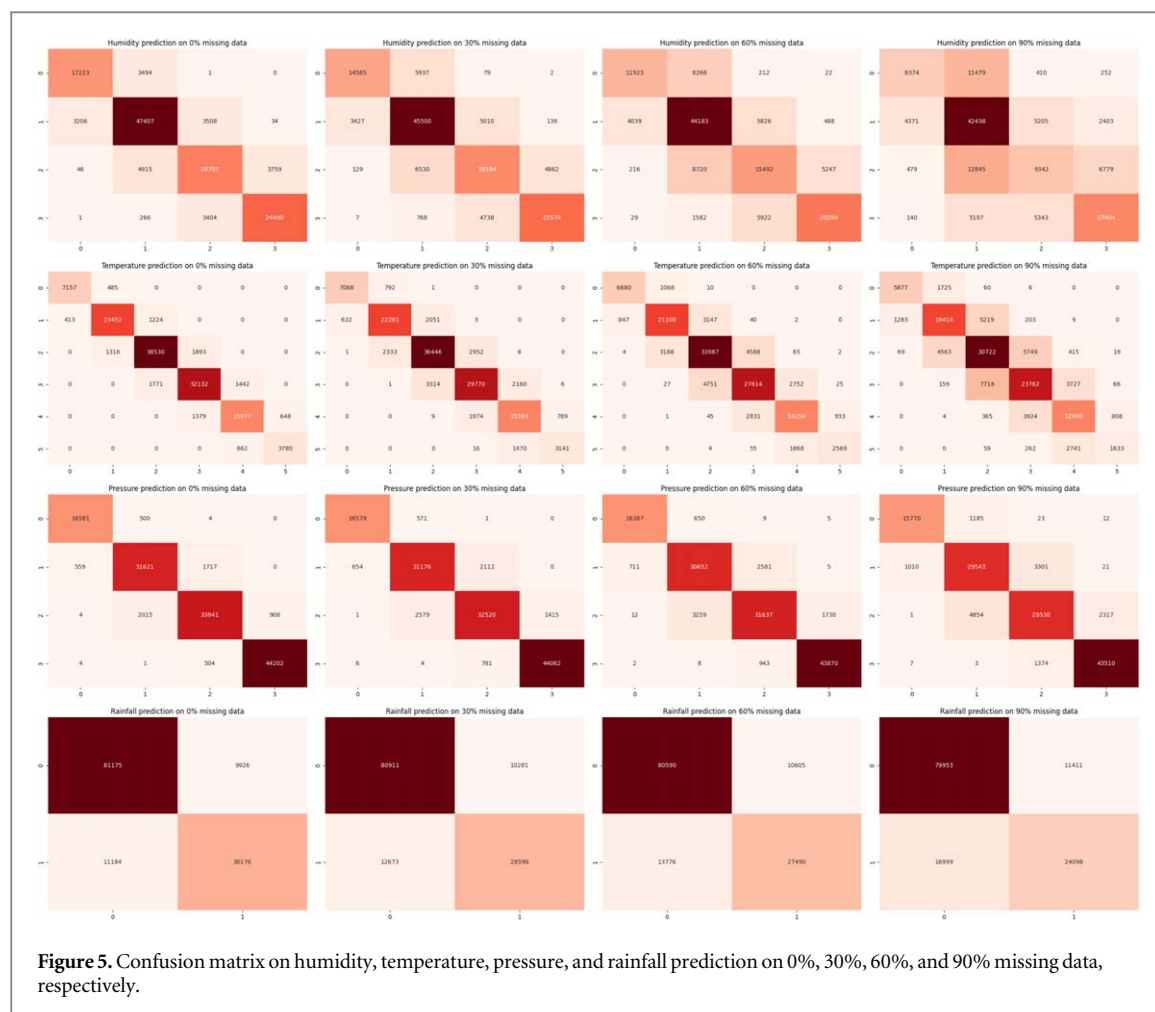
where $W_l \cdot \text{AGG}(\{h_u^{l-1}, \forall u \in \mathcal{N}(v)\})$ is the neighbor embedding and $B_l h_v^{l-1}$ is the self embedding of node v . Here, W_l and B_l are the learnable parameters that controls how much information from neighboring nodes and node v itself should be taken for lower dimensional representation for the node v . Moreover, AGG is aggregation function given by

$$\text{AGG} = \sum_{u \in \mathcal{N}(v)} \frac{h_u^{l-1}}{|\mathcal{N}(v)|} \quad (7)$$

figure 3 illustrates the process of sampling and aggregation by message passing scheme of GraphSAGE convolutional neural network. Our approach begins by constructing a spatio-temporal graph topology, capturing intricate relationships over time. In the subsequent phase, we systematically assign diverse node features, each represented by a unique color, tailored to specific prediction tasks.

5. Experiments and results

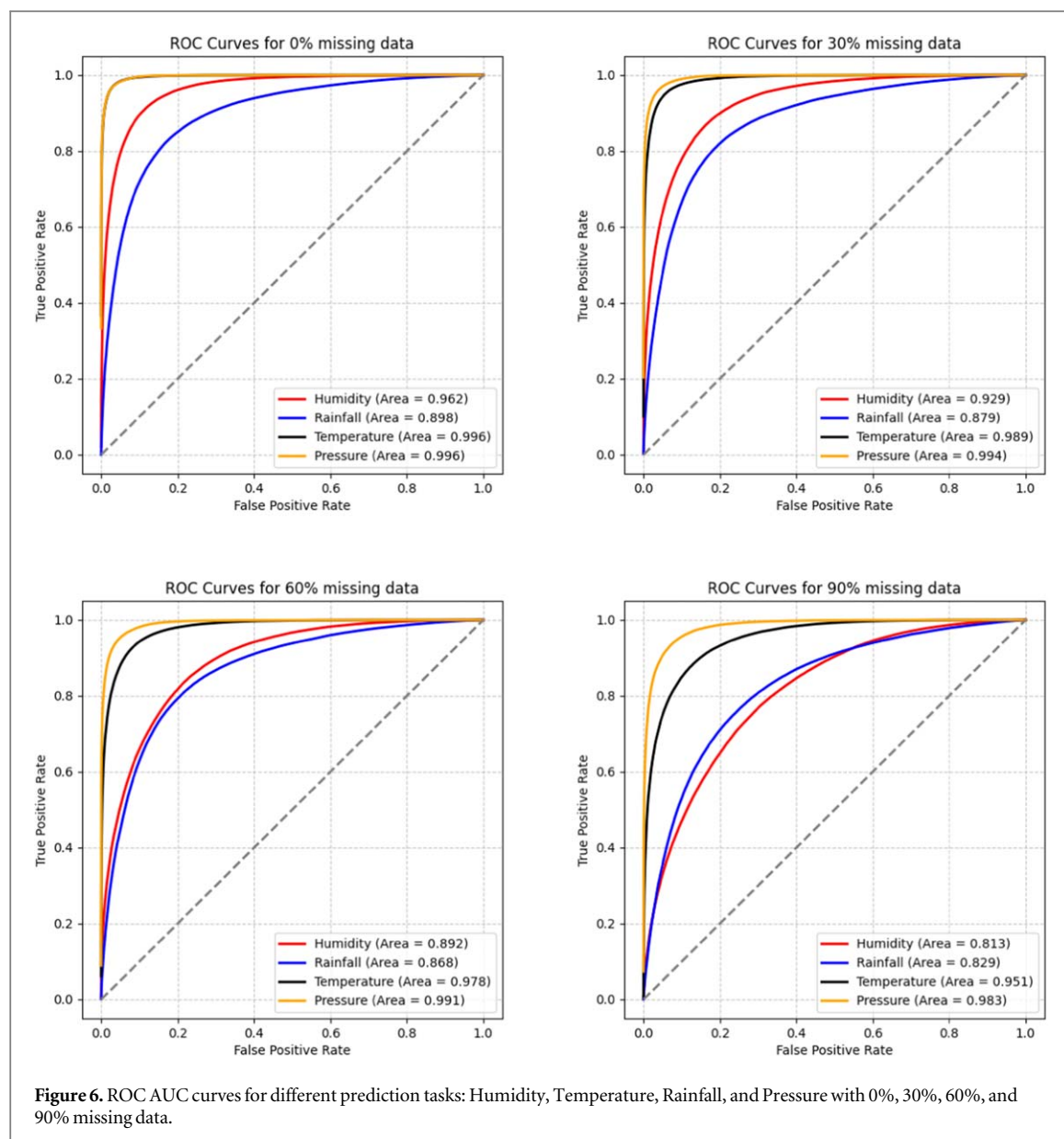
In order to ensure the comprehensibility and comparability of our research findings across different research groups, we consistently present our experimental data in both visual and statistical formats. In this study, we employed GCNs for imputing missing weather data and identifying absent atmospheric variables within a specific range. This was accomplished by leveraging available meteorological data. We utilized several GNN classifiers to predict missing variables, including temperature, humidity, air pressure, and rainfall. In the experiment, we employed Pearson's correlation coefficients for feature selection, optimizing the identification of these four crucial meteorological parameters. Our investigation involved testing our proposed system across various missing data rates, even up to 90%, and the results demonstrated a high classification score. To quantitatively and qualitatively validate our findings, we employed a range of metrics [48], including accuracy, F1-score, confusion matrix, Receiver Operating Characteristic (ROC) score, and t-SNE (t-Distributed Stochastic Neighbor Embedding) plots. The confusion matrix proved valuable for reporting results in multi-class



classification scenarios. Additionally, the area under the ROC curve (AUC) helped assess the degree of separability across multiple classification thresholds. Finally, t-SNE emerged as a valuable tool for data exploration, visualization, informed decision-making, enhanced model performance, and effective communication of our research outcomes.

An ideal imputation method is one that effectively learns to restore the original relationships within a dataset [49]. As discussed in section 1, several methods are proposed over time to achieve the ideal imputation and graph-based imputation approach is found more efficient for temporal data [50]. A graph-based representation to the temporal context is crucial to maintain temporal dynamics, ensuring that the resulting representation preserves the inherent relationships between features and labels over time [51]. Utilizing the strength of graph, we adopted a graph-based feature imputation approach using a diffusion scheme. In this process, we allow a maximum of 30 iterations for the features to propagate throughout the dataset. Following the imputation phase, we employ a three-layer Graph Neural Network (GNN) model. We trained the GNN classifiers; described in section 4; using a large meteorological data of approximately 39 years sourced from 62 weather stations located in Nepal. During training, an early stopping with a patience of 200 epochs was implemented to mitigate overfitting. To construct the graph dataset and leverage graph neural networks, we utilized the PyG (PyTorch Geometric) library, accessible at <https://pytorch-geometric.readthedocs.io/>. The model training and inference were executed on hardware equipped with an NVIDIA GeForce RTX 3080Ti and an Intel i7-11700k@3.60 GHz processor. In this experiment, we evaluated and compared the performance of various graph network architectures, such as GCN, GAT, SAGE, and SAGE-GAT. The results of our experiments as illustrated in Table 1, focusing on humidity, temperature, rainfall, and pressure prediction under different levels of missing data, are illustrated in figure 4. Remarkably, GraphSAGE consistently exhibits superior performance compared to the other classifiers in most test cases. As a result, we focus our subsequent experimental analysis on this superior classifier.

The confusion matrix included in figure 5 also support our hypothesis that the proposed method is robust to the missing data imputation. The GraphSAGE classifier demonstrates robustness in predicting temperature, even when dealing with a large number of classes. However, its performance is suboptimal when predicting rainfall, despite the binary classification task of 'Rainfall' or 'not-Rainfall.' Across all atmospheric variable range



classifications, the classifier exhibits notable confusion, particularly among adjacent classes. For example, in humidity classification, it tends to confuse ‘dry’ and ‘moderate’ classes, ‘moderate’ and ‘humid’ classes, as well as ‘humid’ and ‘very humid’ classes. While some level of confusion is expected, the results for rainfall prediction are relatively poor, indicating that predicting rainfall is one of the most challenging aspects of weather prediction.

The ROC curve in figure 6 presents the macro-average of ROC-AUC scores across all classes while considering different data availability scenarios, which include full data availability (100%), 70% data availability, 40% data availability, and a minimal 10% data availability. Our research findings indicate that the ROC-AUC values remain notably high, even when only 10% of the data is accessible, demonstrating the robustness and effectiveness of the proposed model.

6. Conclusion

As mentioned in the introduction, existing methods primarily focus on grid-based structures, which struggle with sparse weather station data, and one of the widely used NWP methods is heavily influenced by initial data, making it unsuitable for simultaneous spatial and temporal data gaps. Additionally, these methods do not account for dependent weather variables when imputing missing data. Our method, which integrates graph diffusion to handle randomly missing independent variables and utilizes Graph Neural Networks (GNNs) to forecast the ranges of key weather attributes like temperature, rainfall, humidity, and pressure, presents a noteworthy advancement in meteorological modeling tailored for data-sparse environments. By introducing a

novel feature selection approach based on correlation coefficients, we further refine the precision and relevance of our methodology in weather data analysis. Extensively trained on a comprehensive and diverse meteorological dataset encompassing various altitudes and geographic locations, our model demonstrates robustness. Significantly, it achieves remarkable evaluation scores both in the presence and absence of weather attributes, validating the effectiveness and versatility of our approach. Thorough validation through statistical analyses and graphical assessments, employing a range of tailored metrics for classification problems, strengthens the credibility of our method and signifies a substantial step forward in improving the accuracy and reliability of weather attribute predictions in data-sparse regions.

Data availability statement

No new data were created or analysed in this study.

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References

- [1] Talib M N A, Ahmed M, Naseer M M, Slusarczyk B and Popp J 2021 The long-run impacts of temperature and rainfall on agricultural growth in sub-saharan africa *Sustainability* **13** 595
- [2] Zemicheal T and Dietterich T G 2019 Anomaly detection in the presence of missing values for weather data quality control in *Proceedings of the 2nd ACM SIGCAS Conference on Computing and Sustainable Societies COMPASS 2019: ACM SIGCAS Conference on Computing and Sustainable Societies Accra Ghana July 3 - 5, 2019* 65–73
- [3] Kim T, Ko W and Kim J 2019 Analysis and impact evaluation of missing data imputation in day-ahead pv generation forecasting *Appl. Sci.* **9** 204
- [4] Kapoor R, Gupta R, Son L H, Kumar R and Jha S 2019 Fog removal in images using improved dark channel prior and contrast limited adaptive histogram equalization *Multimedia Tools and Applications* **78** 281–23
- [5] Shakya S R and Jha S 2022 Challenges in industrial internet of things (iiot) in *Industrial Internet of Things* (CRC Press) 19–39
- [6] Parasyris A, Alexandrakis G, Kozyrakis G V, Spanoudaki K and Kampanis N A 2022 Predicting meteorological variables on local level with sarima, lstm and hybrid techniques *Atmosphere* **13** 878
- [7] Li J, Wu P, Guo H, Li R, Li G and Xu L 2023 Multivariate transfer passenger flow forecasting with data imputation by joint deep learning and matrix factorization *Applied Sciences* **13** 5625
- [8] Li P, Yu Y, Huang D, Wang Z-H and Sharma A 2023 Regional heatwave prediction using graph neural network and weather station data *Geophysical Research Letters* **50** e2023GL103405
- [9] Tran T T K, Bateni S M, Ki S J and Vosoughifar H 2021 A review of neural networks for air temperature forecasting *Water* **13** 1294
- [10] Tao H, Jawad A H, Shather A, Al-Khafaji Z, Rashid T A, Ali M, Al-Ansari N, Marhoon H A, Shahid S and Yaseen Z M 2023 Machine learning algorithms for high-resolution prediction of spatiotemporal distribution of air pollution from meteorological and soil parameters *Environment international* **175** 107931
- [11] Espeholt L et al 2022 Deep learning for twelve hour precipitation forecasts *Nature communications* **13** 1–10
- [12] Khan M I and Maity R 2020 Hybrid deep learning approach for multi-step-ahead daily rainfall prediction using gcm simulations *IEEE Access* **8** 774–52
- [13] Barrera-Animas A Y, Oyedele L O, Bilal M, Akinosho T D, Delgado J M D and Akanbi L A 2022 Rainfall prediction: A comparative analysis of modern machine learning algorithms for time-series forecasting *Machine Learning with Applications* **7** 100204
- [14] Yen M-H, Liu D-W, Hsin Y-C, Lin C-E and Chen C-C 2019 Application of the deep learning for the prediction of rainfall in southern taiwan *Scientific reports* **9** 12774
- [15] Wu Z, Pan S, Long G, Jiang J, Chang X and Zhang C 2020 Connecting the dots: Multivariate time series forecasting with graph neural networks in *Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining* 753–63
- [16] Bai L, Yao L, Li C, Wang X and Wang C 2020 Adaptive graph convolutional recurrent network for traffic forecasting *Advances in neural information processing systems* **33** 17804–15
- [17] Lira H, Martí L and Sanchez-Pi N 2022 A graph neural network with spatio-temporal attention for multi-sources time series data: An application to frost forecast *Sensors* **22** 1486
- [18] Ma M, Xie P, Teng F, Wang B, Ji S, Zhang J and Li T 2022 Available at SSRN 4455568 *Histgmn: Hierarchical spatio-temporal graph neural network for weather forecasting* **648** 119580
- [19] Park I, Kim H S, Lee J, Kim J H, Song C H and Kim H K 2019 Temperature prediction using the missing data refinement model based on a long short-term memory neural network *Atmosphere* **10** 718
- [20] Afrifa-Yamoah E, Mueller U A, Taylor S and Fisher A 2020 Missing data imputation of high-resolution temporal climate time series data *Meteorological Applications* **27** e1873
- [21] Zuo J, Zeitouni K, Taher Y and Garcia-Rodriguez S 2023 Graph convolutional networks for traffic forecasting with missing values *Data Mining and Knowledge Discovery* **37** 913–47
- [22] Kim J-W and Pachepsky Y A 2010 Reconstructing missing daily precipitation data using regression trees and artificial neural networks for swat streamflow simulation *Journal of hydrology* **394** 305–14
- [23] Nourani V, Baghanam A H and Gebremichael M 2012 Investigating the ability of artificial neural network (ann) models to estimate missing rain-gauge data *Journal of Environmental Informatics* **19**
- [24] Teegavarapu R S 2014 Statistical corrections of spatially interpolated missing precipitation data estimates *Hydrological Processes* **28** 3789–808

- [25] Ferrari G T and Ozaki V 2014 Missing data imputation of climate datasets: Implications to modeling extreme drought events *Revista Brasileira de Meteorologia* **29** 21–8
- [26] Alsaber A R, Pan J and Al-Hurban A 2021 Handling complex missing data using random forest approach for an air quality monitoring dataset: a case study of kuwait environmental data (2012 to 2018) *International Journal of Environmental Research and Public Health* **18** 1333
- [27] Jordanov I, Petrov N and Petrozziello A 2018 Classifiers accuracy improvement based on missing data imputation *Journal of Artificial Intelligence and Soft Computing Research* **8** 31–48
- [28] Xie C, Huang C, Zhang D and He W 2021 Bilstm-i: A deep learning-based long interval gap-filling method for meteorological observation data *International Journal of Environmental Research and Public Health* **18** 10321
- [29] Mital U, Dwivedi D, Brown J B, Faybishenko B, Painter S L and Steefel C I 2020 Sequential imputation of missing spatio-temporal precipitation data using random forests *Frontiers in Water* **2** 20
- [30] Sanhudo L, Rodrigues J and Filho E V 2021 Multivariate time series clustering and forecasting for building energy analysis: Application to weather data quality control *Journal of Building Engineering* **35** 101996
- [31] Guastella D A, Marcillaud G and Valenti C 2021 Edge-based missing data imputation in large-scale environments *Information* **12** 195
- [32] Bauer P, Thorpe A and Brunet G 2015 The quiet revolution of numerical weather prediction *Nature* **525** 47–55
- [33] Morales Martínez J L, Horta-Rangel F A, Segovia-Domínguez I, Robles Morua A and Hernández J H 2019 Analysis of a new spatial interpolation weighting method to estimate missing data applied to rainfall records *Atm ósfera* **32** 237–59
- [34] Al-Yahyai S, Charabi Y and Gastli A 2010 Review of the use of numerical weather prediction (nwp) models for wind energy assessment *Renewable and Sustainable Energy Reviews* **14** 3192–8
- [35] <https://opendatanepal.com/dataset/district-wise-daily-climate-data-for-nepal>
- [36] NASA 2023 *Nasa power data access viewer*
- [37] Bu J, Peng D-D, Xiao H, Yue Q, Han Y, Lin Y, Hu G and Chen J 2020 Analysis of meteorological conditions and prediction of epidemic trend of 2019-ncov infection in 2020 *MedRxiv* 2020–2
- [38] Islam M A, Sangkham S, Tiwari A, Vadiati M, Hasan M N, Noor S T A, Mumin J, Bhattacharya P and Sherchan S P 2022 Association between global monkeypox cases and meteorological factors *International Journal of Environmental Research and Public Health* **19** 15638
- [39] Lal M, Sharma S, Yadav S and Kumar S 2018 Management of late blight of potato *Potato-From Incas to All Over the World* 83–106
- [40] Rajagopal A, Jha S, Khari M, Ahmad S, Alouffi B and Alharbi A 2021 A novel approach in prediction of crop production using recurrent cuckoo search optimization neural networks *Applied Sciences* **11** 9816
- [41] Jha S, Nkenyereye L, Joshi G P and Yang E 2020 Mitigating and monitoring smart city using internet of things *Computers, Materials & Continua* **65** 1059–79
- [42] Jha S, Joshi G P, Nkenyereye L, Kim D W and Smarandache F 2020 A direct data-cluster analysis method based on neutrosophic set implication *Infinite Study* **65** 1203–1220
- [43] Rossi E, Kenlay H, Gorinova M I, Chamberlain B P, Dong X and Bronstein M M 2022 On the unreasonable effectiveness of feature propagation in learning on graphs with missing node features in *Learning on Graphs Conference* 11–1
- [44] Seo J-H et al 2014 Feature selection for very short-term heavy rainfall prediction using evolutionary computation *Advances in Meteorology* **2014**
- [45] Malathi T and Manimekalai M 2020 Feature selection techniques for weather forecasting models using machine learning techniques *Journal of Electrical Engineering and Technology* **11** 443–55
- [46] Moon S-H and Kim Y-H 2020 An improved forecast of precipitation type using correlation-based feature selection and multinomial logistic regression *Atmospheric Research* **240** 104928
- [47] Hamilton W, Ying Z and Leskovec J 2017 Inductive representation learning on large graphs *Advances in neural information processing systems* **30**
- [48] Singh D et al 2023 Lumbar spine disease detection: Enhanced cnn model with improved classification accuracy *IEEE Access* **11** 141889–141901
- [49] Yoon J 2020 University of California *End-to-end machine learning frameworks for medicine: data imputation, model interpretation and synthetic data generation* (ProQuest LLC)
- [50] Gordon D, Petousis P, Zheng H, Zamanzadeh D and Bui A A 2021 Tsi-gnn: Extending graph neural networks to handle missing data in temporal settings *Frontiers in big Data* **4** 693869
- [51] Meng X-L 1994 Multiple-imputation inferences with uncongenial sources of input *Statistical science* **9** 538–58